

ASSIGNMENT

- 1 Find the number of ways in which 4 boys and 2 girls (all are of different heights) can be arranged in a line so that boys as well as girls among themselves are in decreasing order of height (from left to right).

Solution:

$${}^6C_2 \times 1 \times 1 = 15$$

- 2 The number of combination of 16 things, 8 of which are alike and the rest different, taken 8 at a time is

Solution:

As selecting 8 objects from n objects which are alike is 1

Number of ways

$$\begin{aligned} &= {}^8C_0 + {}^8C_1 + {}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8 \\ &= 2^8 = 256 \end{aligned}$$

- 3 In how many ways we can pick 5 letters out of ARRANGEMENT

Solution:

a) Given, 2A, 2R, 2N, 2E, 1M, 1T, 1G

Different 5 letter combination can be

$$\begin{aligned} \text{(i) 2 alike, 2 alike, 1 different} &= {}^4C_2 \times {}^5C_1 \\ &= 6 \times 5 = 30 \end{aligned}$$

$$\begin{aligned} \text{(ii) 2 alike, 3 different} &= {}^4C_1 \times {}^6C_3 \\ &= 4 \times 20 = 80 \end{aligned}$$

$$\text{(iii) All different} = {}^7C_5 = 21$$

$$\therefore \text{total number of ways} = 30 + 80 + 21 = 131$$

- 4 The interior angles of a regular polygon measure 150° each. Find the number of diagonals of the polygon.

Solution:

Let n be the number of sides of a polygon

$$\therefore \frac{(n-2)180^\circ}{n} = 150^\circ$$

$$\Rightarrow (n-2)6 = 5n$$

$$\Rightarrow 6n - 5n = 12$$

$$\Rightarrow n = 12$$

$$\therefore \text{Number of diagonal} = {}^{12}C_2 - 12$$

$$= 66 - 12 = 54$$

- 5 18 points are indicated on the perimeter of a triangle ABC (see figure). How many triangles are there with vertices at these points?



Solution:

There are total 18 points

Out of which 7, 7, 7 are respective collinear points on sides

$\therefore$ Total number of triangles

$$= {}^{18}C_3 - 3({}^7C_3)$$

$$= 711$$

- 6 Consider 9 points out of which 4 are collinear. Joining these points how many

Option:

(a) lines will be made

(b) triangles will be made

(c) Quadrilateral will be made

(d) Pentagons will be made

Answer: (c)

Solution:

(a) Given out of 9 points, 5 are non-collinear and 4 are collinear

$$\therefore \text{Number of lines} = {}^9C_2 - {}^4C_2 + 1$$

$$= 36 - 6 + 1 = 31$$

(b) Triangles can be made

$$\begin{aligned}
 &= {}^5C_3 + {}^5C_2 \times {}^4C_1 + {}^5C_1 \times {}^4C_2 \\
 &= 10 + 40 + 30 \\
 &= 80
 \end{aligned}$$

(c) quadrilateral can be made

$$\begin{aligned}
 &= {}^5C_4 + {}^5C_3 \times {}^4C_1 + {}^5C_2 \times {}^4C_2 \\
 &= 5 + 40 + 60 \\
 &= 105
 \end{aligned}$$

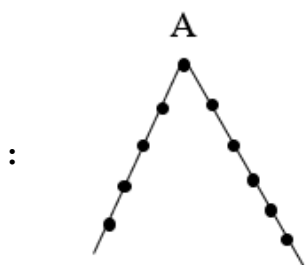
(d) Pentagons can be made

$$\begin{aligned}
 &= {}^5C_5 + {}^5C_4 \times {}^4C_1 + {}^5C_3 \times {}^4C_2 \\
 &= 1 + 20 + 60 \\
 &= 81
 \end{aligned}$$

7 Triangles are formed joining the marked points as vertices, when

(i) A is excluded &

(ii) A is included Find ratio of number of triangles



Solution:

(i) when A is excluded then

Number of triangles

$$\begin{aligned}
 &= {}^9C_3 - {}^4C_3 - {}^5C_3 \\
 &= 84 - 4 - 10 \\
 &= 70
 \end{aligned}$$

(ii) when A is included, then

Number of triangles

$$= {}^{10}C_3 - {}^5C_3 - {}^6C_3$$

$$= 90$$

$$\therefore \text{ratio} = \frac{70}{90} = 7:9$$

- 8 There are m points on a straight line AB and n points on the line AC none of them being the point A . Triangles are formed with these points as vertices, when
- (i) A is excluded
- (ii) A is included

The ratio of number of triangle in the two cases is:

Solution:

Generalizing, given m points on AB and n points on AC (excluding A)

(i) Number of triangles when A is excluded

$$= {}^{m+n}C_3 - {}^mC_3 - {}^nC_3$$

$$= m + n - 2$$

(ii) Number of triangles when A is included

$$= {}^{m+n+1}C_3 - {}^{m+1}C_3 - {}^{n+1}C_3$$

$$= m + n$$

$$\therefore \text{ratio} = \frac{m + n - 2}{m + n}$$

- 9 In a plane there are 6 lines, no two of which are parallel & no three are concurrent. How many triangles can be formed with their points of intersection as vertices

Solution:

Here, 6 lines will give point of intersection = ${}^6C_2 = 15$

Also, all other 5 lines intersect collinearly on one line such that those 5 points are collinear

$\therefore$ Number of triangles

$$= {}^{15}C_3 - 6({}^5C_3)$$

$$= 455 - 6(10)$$

$$= 455 - 60$$

$$= 395$$

- 10 How many squares & rectangles are there in 8×8 chessboard

Solution:

In a chess board there are 9 vertical lines and 9 horizontal lines

$$\therefore \text{Number of rectangles} = {}^9C_2 \times {}^9C_2 = 1296$$

For squares, we have to take lines at equal distance

Means 1-2, 2-3, 3-4, ..., 8-9 (8 horizontal and 8 vertical)

or 1-3, 2-4, ..., 7-9 (7 horizontal and 7 vertical)

and so on

$\therefore$ Number of squares

$$= 8^2 + 7^2 + 6^2 + \dots + 2^2 + 1^2$$

$$= \frac{8(8+1)(16+1)}{6}$$

$$= 204$$

- 11** (a) Out of seven consonants and four vowels, the number of words of six letters, formed by taking four consonants and two vowels is (Assume that each ordered group of letter is a word):
 (b) Find the number of permutations of the word "AUROBIND" in which vowels appear in alphabetical order.
 (c) Find the number of permutations of the word "AUROBIND" in which no two vowels come together.

Solution:

(a) Six letters words containing 4 consonants out of given 7 consonants and 2 vowels out of given 4 vowels

$$= {}^7C_4 \times {}^4C_2 \times 6! \\ = 151200$$

(b) Arrangements in which vowels appear in alphabetical order = $\frac{8!}{4!}$

(c) ${}^5C_4 4!4!$

- 12** Find the number of ways in which letters of the word VALEDICTORY be arranged so that the vowels may never be separated.

Solution:

Given, V, L, D, C, T, R, Y, A, E, I, O

Keeping vowels together

Number of different words = $8! \times 4! = 967680$

- 13 Four visitors A, B, C & D arrive at a town which has 5 hotels. In how many ways can they disperse themselves among 5 hotels, if 4 hotels are used to accommodate them

Solution:

Out of 5 hotels only 4 are used to accommodate

$\therefore$ Number of ways = ${}^5C_4 \times 4!$

= $5 \times 24 = 120$ ways

- 14 Number of six digit numbers which have 3 digits even and 3 digits odd, if each is to be used atmost once is

Solution:

Here, we have 5 odd digits and 5 even digits which includes 0 also which can't be in the first position

$\therefore$ Total such number

$$= {}^5C_3 \times {}^5C_3 \times 6! - {}^5C_3 \times {}^4C_2 \times 5!$$

$$= 72000 - 7200$$

$$= 64800$$

- 15 If m denotes the number of 5 digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then (m - n) has the value

Solution:

Here, m denotes 5 digit number in descending order = ${}^{10}C_5$

n denotes 5 digit number in ascending order = 9C_5

$$\therefore m - n = {}^{10}C_5 - {}^9C_5 = {}^9C_5$$

- 16 There are n concurrent lines and another line parallel to one of them. The number of different triangles that will be formed by the (n + 1) lines, is

Solution:

In this case, one side will be the parallel line which is non-concurrent.

The other parallel line cannot form any triangle.

So, the remaining two sides can be chosen in ${}^{n-1}C_2$ ways

$$= \frac{(n-1)(n-2)}{2}$$

- 17 The number of ways in which we can choose 3 squares of unit area on a chess board such that one of the squares has its two sides common to other two squares

Solution:

Total number of square having unit area on chess board = $8 \times 8 = 64$

1 1'

2 2'

3 3'

For square at 1 position, No of ways of choosing 3 square = 1

For square at 1' position, No of ways of choosing 3 square = 3

For square at 2 position, No of ways of choosing 3 square = 3

At position 2, Number of ways of choosing 3 square = 6,

So, Total number of ways of choosing 3 square

$$= \underbrace{(2 \times 1 + 6 \times 3) \times 2}_{\text{upper and lower row}} + \underbrace{(3 \times 2 + 6 \times 6) \times 6}_{\text{middle row}}$$

$$= 40 + (42) \times 6$$

$$= 292$$

- 18** Given 11 points, of which 5 lie on one circle, other than these 5, no 4 lie on one circle. Then the maximum number of circles that can be drawn so that each contains at least three of the given points is

Solution:

Given that, In 11 points, 5 lie on one circle, other than these, no 4 lie on one circle.

As we know, from 3 points one circle will be formed.

So,

i) Selecting 2 points from 5 lie on one circle and 1 from other circle. = ${}^5C_2 \times {}^6C_1$
= 60

ii) Selecting 1 points from 5 lie on one circle and 2 from other circle = ${}^5C_1 \times {}^6C_2$
= 75

iii) Selecting all 3 points from other 6 points = 6C_3

iv) One circle is already drawn from 5 points

Total number of circles = $60 + 75 + 20 + 1 = 156$

- 19** Two classrooms A and B having capacity of 25 and $(n - 25)$ seats respectively. A_n denotes the number of possible seating arrangements of room 'A', when 'n' students are to be seated in these rooms, starting from room 'A' which is to be filled up full to its capacity.

If $A_n - A_{n-1} = 25!$ (${}^{49}C_{25}$) then 'n' equals

Solution:

We can clearly assume $n > 25$

$$A_n = {}^nC_{25} \cdot 25!$$

[selecting 25 student from n students and arranging then in 25 positions]

$$A_{n-1} = {}^{n-1}C_{25} \cdot 25!$$

We have

$$A_n - A_{n-1} = 25! \cdot {}^{49}C_{25}$$

$${}^nC_{25} \cdot 25! - {}^{n-1}C_{25} \cdot 25! = 25! \cdot {}^{49}C_{25}$$

$${}^nC_{25} - {}^{n-1}C_{25} = {}^{49}C_{25}$$

$$\frac{n!}{(n-25)!25!} - \frac{(n-1)!}{(n-26)!25!} = \frac{49!}{25!24!}$$

$$\frac{(n-1)!}{(n-26)!} \left[\frac{n}{n-25} - 1 \right] = \frac{49!}{24!}$$

$$\frac{(n-1)!}{(n-26)!} \times \frac{25}{(n-25)} = \frac{49!}{24!}$$

$$\frac{(n-1)!}{(n-25)!} = \frac{49!}{25!}$$

$$n-1 = 49$$

$$n = 50$$

- 20 In how many ways it is possible to select six letters, including at least one vowel from the letters of the word "FLABELLIFORM". (It is a picnic spot in U. S. A.)

Solution:

FLABEIORM

23

Consider all cases-

$$\text{All 6 different} = {}^9C_6 = 84$$

$$2 \text{ same } 4 \text{ different} = {}^2C_1 \times {}^8C_4 = 140$$

$$2 \text{ same } 2 \text{ same } 2 \text{ different} = {}^7C_2 = 21$$

$$3 \text{ same } 2 \text{ same } 1 \text{ different} = {}^7C_1 = 7$$

$$3 \text{ same } 3 \text{ different} = {}^8C_3 = \frac{56}{308}$$

Now, not considering vowels-

$$\text{All 6 different} \rightarrow {}^6C_6 = 1$$

$$2 \text{ same } 4 \text{ different} \rightarrow {}^2C_1 \times {}^5C_4 = 10$$

$$2 \text{ same } 2 \text{ same } 2 \text{ different} \rightarrow {}^4C_2 = 6$$

$$3 \text{ same } 2 \text{ same } 1 \text{ different} \rightarrow {}^4C_1 = 4$$

$$3 \text{ same } 3 \text{ different} \rightarrow {}^5C_3 = \frac{10}{31}$$

Total favorable cases = 271